

# Probing Large Language Models for Autonomous Driving Behavior

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## Appendix

### A1. Evaluation of Translation Artifacts in Multilingual Prompts

Because the main analysis identifies systematic differences across English, Chinese, and French prompts, it is important to verify that these differences are not merely caused by translation artifacts. To address this concern, we conducted an additional back-translation evaluation on the multilingual prompt sets used in the main experiments. The purpose of this appendix is to examine whether the non-English prompts preserve the semantic content, and tone and framing of the original English prompts with sufficient fidelity. The evaluation combines two complementary analyses: a semantic similarity analysis based on automated metrics, and a tone/framing analysis based on an LLM-as-judge protocol. Together, these analyses assess whether translation quality is sufficiently high to rule out prompt inconsistency as a plausible confounding explanation for the cross-language effects reported in the main text.

#### A1.1 Back-Translation Protocol

The multilingual experiments in the main paper are conducted in four driving scenarios: handling pedestrian crossings, following a cyclist, yielding to a merging vehicle, and merging into freeway. For each scenario, the Chinese and French prompt sets are translated back into English using the Gemini 3 (Team et al., 2023) which is different from the one (GPT4o) that generates the corresponding forward translations. This procedure yields eight back-translated prompt sets in total, corresponding to four scenarios under two translation directions (Chinese to English and French to English). Each back-translated prompt is then compared against its corresponding original English prompt. Consistent with the structure of the experimental prompts, the comparison is conducted at two levels: the system prompt, which specifies the task instructions and scenario context, and the behavioral options, which define the action space available to the model.

#### A1.2 Semantic Similarity Analysis

To quantify semantic preservation, we employ three complementary metrics that capture similarity at different linguistic levels: embedding cosine similarity, BERTScore (Zhang et al., 2019), and METEOR (Banerjee & Lavie, 2005). These metrics are computed for the system prompt alone,

and behavioral options prompt individually. Since the behavioral options directly define the ordered driving actions used in the main experiments, preserving their meaning is especially important for ruling out translation-induced shifts in decision space.

First, we use embedding cosine similarity based on the text-embedding-3-large model to measure global semantic alignment between the original English text and its back-translated version. Let  $e_o$  and  $e_b$  denote the embedding vectors of the original and back-translated texts, respectively. Their semantic similarity is defined as

$$\text{CosSim}(e_o, e_b) = \frac{e_o^T e_b}{\|e_o\| \|e_b\|}. \quad (\text{A1})$$

Second, we use BERTScore to measure token-level semantic preservation based on contextual embeddings. Denoting the precision and recall components of BERTScore as  $P_{\text{BERT}}$  and  $R_{\text{BERT}}$ , the reported F1 score is

$$\text{BERTScore}_{F1} = 2 \frac{P_{\text{BERT}} R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}. \quad (\text{A2})$$

Compared with embedding cosine similarity, BERTScore is more sensitive to local semantic correspondence and is therefore useful for detecting subtle distortions in option descriptions or instruction wording. Third, we use METEOR to evaluate lexical-semantic alignment while allowing for stemming and synonym matching. This metric is suitable for back-translation assessment because semantically faithful paraphrases may differ in surface form. In this setting, reformulations such as “reduce speed” versus “decelerate” should not be interpreted as evidence of semantic inconsistency. Let  $P$  and  $R$  denote unigram precision and recall, respectively, computed after exact matching, stem matching, and synonym matching. Their weighted harmonic mean is defined as

$$F_{\text{mean}} = \frac{PR}{\alpha P + (1-\alpha)R}, \quad (\text{A3})$$

where  $\alpha$  controls the relative weight assigned to precision and recall. To account for word order, METEOR further introduces a fragmentation penalty based on the number of matched chunks. Let  $m$  denote the number of matched unigrams and  $ch$  the number of matched chunks. The penalty term is defined as

$$\text{Pen} = \gamma \left( \frac{ch}{m} \right)^\beta, \quad (\text{A4})$$

where  $\gamma$  and  $\beta$  are tunable parameters. The final METEOR score is then computed as

$$\text{METEOR} = (1 - \text{Pen}) \cdot F_{\text{mean}}. \quad (\text{A5})$$

**Table A1** Summary of semantic similarity metrics for back-translated prompts.

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| Translation direction | Prompt component   | Cosine similarity (mean) | Cosine similarity (min) | BERTScore F1 (mean) | BERTScore F1 (min) | METEOR (mean) |
|-----------------------|--------------------|--------------------------|-------------------------|---------------------|--------------------|---------------|
| Chinese → English     | System prompts     | 0.969                    | 0.963                   | 0.973               | 0.970              | 0.789         |
| Chinese → English     | Behavioral options | 0.926                    | 0.860                   | 0.960               | 0.942              | 0.749         |
| French → English      | System prompts     | 0.979                    | 0.977                   | 0.975               | 0.970              | 0.867         |
| French → English      | Behavioral options | 0.918                    | 0.835                   | 0.972               | 0.951              | 0.827         |

**Table A1** summarizes the semantic similarity results by translation direction and prompt component. The results indicate consistently high semantic preservation across both translation directions. At the system-prompt level, cosine similarity remains above 0.96 in all cases, showing that the core task framing and scenario instructions are preserved with very high fidelity. At the behavioral-option level, the mean cosine similarity remains above 0.91, while the minimum values of 0.860 and 0.835 still correspond to high BERTScore F1 values. Importantly, all BERTScore F1 values exceed 0.94, with mean values ranging from 0.960 to 0.975. This indicates that even when lexical reformulation occurs, the token-level semantic content remains highly consistent. METEOR values are lower than cosine similarity and BERTScore, but this is expected because METEOR is more sensitive to surface-level wording differences and word-order fragmentation (Banerjee & Lavie, 2005). In the present context, these lower values reflect acceptable paraphrastic variation rather than substantive semantic drift. Overall, **Table A1** provides strong evidence that the multilingual prompts preserve the original semantic content, including the behaviorally critical option descriptions used in the experiments.

### A1.3 Tone and Framing Analysis

Although semantic similarity metrics can verify meaning preservation, they do not fully capture whether a translation subtly alters the tone or behavioral framing of a prompt. Such changes could, in principle, affect model decisions even when the core semantics remain intact. We therefore perform an additional tone/framing analysis using an LLM-as-judge protocol, with the judge system prompt provided in **Fig. A1**. For each of the eight comparison pairs, the original English prompt and its back-translated English version are jointly evaluated along six dimensions: command strength, politeness/indirectness, ambiguity, negotiation/interaction framing, and risk framing. Each dimension is

scored on a three-level scale, where 0 indicates no meaningful difference, 1 indicates a minor difference unlikely to affect decisions, and 2 indicates a substantial difference likely to alter model behavior. To improve robustness, each comparison is repeated three times with stochastic decoding, and the final score is determined by majority voting.

LLM-as-Judge Prompt for Tone/Framing Evaluation

Role:

You are an expert evaluator in language consistency for decision-making prompts.

Input:

• Prompt A: original English prompt  
 • Prompt B: back-translated English prompt

Task:

Evaluate whether Prompt B differs from Prompt A in tone or framing in a way that could affect decision-making by a language model.

IMPORTANT:

• Focus ONLY on differences that could influence decisions.  
 • Ignore minor wording differences that do NOT affect meaning or interpretation.  
 • Be strict and conservative: only mark "2" if the difference is likely to change model behavior.

Evaluation Dimensions:

1. Command Strength

Does one prompt sound more directive, forceful, or imperative?

2. Politeness / Indirectness

Does one prompt sound more polite, hedged, or indirect?

3. Ambiguity

Does one prompt introduce more vagueness or interpretation flexibility?

4. Negotiation / Interaction Framing

Does one prompt shift between assertive vs cooperative behavior?

5. Risk Framing

Does one prompt emphasize safety vs efficiency differently?

Scoring Scale:

0 – No difference | 1 – Minor (unlikely to affect decisions) | 2 – Substantial (likely to affect)

Output Format (JSON):

{ "command strength": 0/1/2, "politeness": 0/1/2, "ambiguity": 0/1/2, "negotiation framing": 0/1/2, "risk framing": 0/1/2, "explanation": "..." }

**Fig. A1.** LLM-as-judge prompt for tone/framing evaluation.

**Table A2** Tone and framing evaluation results for back-translated prompts.

| Scenario                      | Translation direction | Command strength | Politeness | Ambiguity | Negotiation framing | Risk framing |
|-------------------------------|-----------------------|------------------|------------|-----------|---------------------|--------------|
| Handling pedestrian crossings | Chinese → English     | 0                | 0          | 1         | 0                   | 1            |
| Handling pedestrian crossings | French → English      | 0                | 0          | 0         | 0                   | 0            |
| Following a cyclist           | Chinese → English     | 0                | 0          | 0         | 0                   | 1            |

|                               |                   |   |   |   |   |   |
|-------------------------------|-------------------|---|---|---|---|---|
| Following a cyclist           | French → English  | 0 | 0 | 1 | 0 | 0 |
| Yielding to a merging vehicle | Chinese → English | 0 | 0 | 0 | 0 | 0 |
| Yielding to a merging vehicle | French → English  | 0 | 0 | 0 | 0 | 1 |
| Merging into freeway          | Chinese → English | 0 | 0 | 0 | 0 | 0 |
| Merging into freeway          | French → English  | 0 | 0 | 0 | 0 | 0 |

**Table A2** presents the aggregated results. Several observations are notable. First, none of the eight comparison pairs received a score of 2 on any dimension. That is, the evaluation found no substantial differences that would plausibly alter model decisions. Second, the dimensions most relevant to the behavioral ordering in this study, command strength, politeness, and negotiation framing, received a score of 0 for all eight cases. This indicates that the directive structure of the prompt and the conservative, neutral, and assertive framing of the behavioral options were preserved consistently across translations. Third, the few non-zero scores were confined to ambiguity and risk framing, and all of these were minor. These cases correspond to slight lexical or stylistic reformulations rather than systematic changes in action meaning or behavioral ordering. For example, in the French-to-English back-translation of the “following a cyclist” scenario, which receives a score of 1 on ambiguity, the judge explicitly states that the prompts are “largely equivalent in directives, tone, and safety emphasis” and that the observed difference is only a “minor ambiguity shift” unlikely to materially change decisions. The explanation further notes that wording variations such as “must” versus “should,” “pass” versus “overtake,” and “assertive” versus “firmer” are not expected to meaningfully alter model behavior. This qualitative evidence is consistent with the aggregated scores in Table A2 and further suggests that the score-1 cases do not indicate meaningful distortions of the original prompt intent. Taken together, these findings provide additional evidence that the observed cross-language differences are unlikely to be driven by translation artifacts.

## A2. Open-Ended Response Analysis for the Pedestrian-Crossing Scenario

In the main experiment, the handling pedestrian crossings scenario exhibits a highly concentrated response distribution, with most model outputs falling into the neutral category. This concentration limits the usefulness of the Ordered Logit Model, because the lack of sufficient variation weakens the model’s ability to estimate factors reliably. However, to rule out the possibility that this concentration is induced by the predefined options in the closed-ended prompt setting, we conduct an additional open-ended analysis for the pedestrian-crossing scenario. In this supplementary experiment, the predefined behavioral options are removed, and each model is asked to generate its own driving action directly from the scenario context. This design allows us to assess whether the concentration in the neutral category remains when the models are no longer guided by an explicitly provided option set.

### A2.1 Experimental Design

We randomly sample 200 cases from the original pedestrian-crossing scenario set. Each sampled case preserves the same contextual information used in the main experiment, including environmental conditions, pedestrian characteristics, ego-vehicle state, and occupant urgency. In contrast to the closed-ended setting, the models are not given any predefined driving options. Instead, they are instructed to analyze the scenario, propose an appropriate driving action in free-form text, and

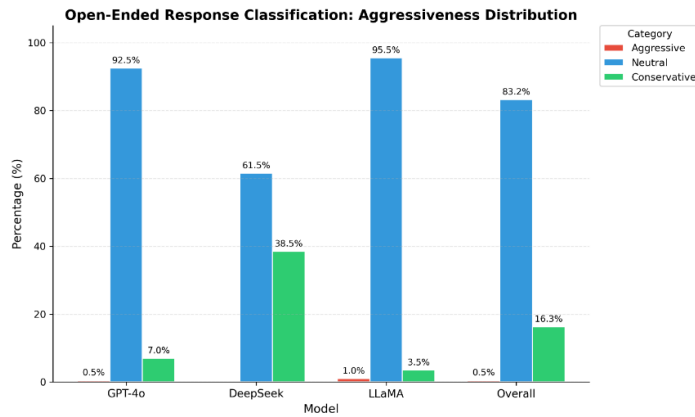
briefly justify that action. This design ensures that the generated responses reflect the models’ intrinsic behavioral preferences rather than their tendency to align with the wording of predefined alternatives. Since the responses are free-form rather than categorical, an additional mapping step is required to compare them with the three-level aggressiveness framework used in the main paper.

### A2.2 Response Mapping Procedure

To map the open-ended responses into the same ordinal framework used in the main experiment, we employ a separate GPT-4o instance as an independent judge. For each generated response, the judge receives the full pedestrian-crossing scenario context together with the model’s proposed action and justification, and assigns the response to exactly one of three categories: Aggressive, Neutral, or Conservative. The classification criteria follow the same conceptual ordering used in the main paper. Specifically, aggressive responses prioritize speed or throughput over pedestrian safety, neutral responses balance caution with continued progress, and conservative responses prioritize pedestrian safety through proactive yielding or near-stopping behavior.

### A2.3 Results

**Fig. A2** illustrates the distribution of aggressiveness categories after the judge maps the open-ended responses to the three behavioral classes. Even after removing predefined options entirely, responses remain overwhelmingly concentrated in the neutral category, with 83.2% of all open-ended responses classified as neutral. Specifically, GPT-4o produces 0.5% aggressive, 92.5% neutral, and 7.0% conservative responses; DeepSeek produces 0.0% aggressive, 61.5% neutral, and 38.5% conservative responses; and LLaMA produces 1.0% aggressive, 95.5% neutral, and 3.5% conservative responses. These proportions are broadly consistent with the findings reported in the main text: GPT-4o and LLaMA remain dominated by neutral responses, whereas DeepSeek, although still primarily neutral, exhibits a larger share of conservative responses. Overall, this consistency between the open-ended and closed-ended settings suggests that the neutral-dominant pattern in the pedestrian-crossing scenario is not mainly induced by predefined option framing, but instead reflects a stable preference for cautious, non-fully-stopping behavior when the AV interacts with a pedestrian near a crosswalk.

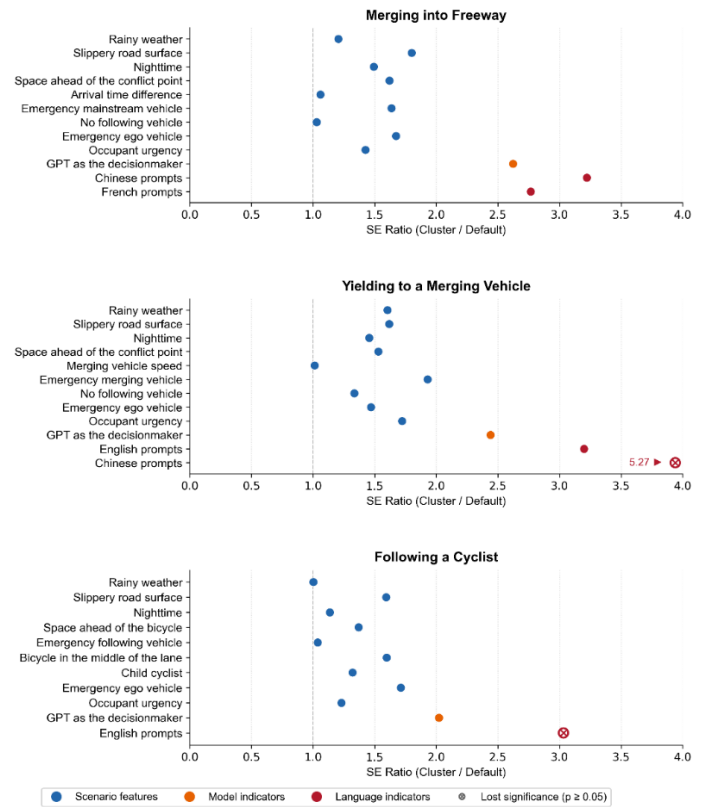


**Fig. A2.** Distribution of aggressiveness categories in the open-ended handling pedestrian crossings scenario experiment.

### A3. Robustness Check: Independence Assumption of the Ordered Logit Model

The Ordered Logit Model (OLM) employed in the main analysis assumes that the error terms are identically and independently distributed across observations (see Eq. 1). Although each observation in this study is generated through an independent API call to the respective LLM, with no shared conversational memory across calls and a temperature setting of 1.0 that introduces stochastic sampling variability, observations within the same model-language group may nonetheless share systematic behavioral tendencies arising from the model's architecture, training data, or language-specific alignment. If such within-group correlation exists, the standard maximum likelihood standard errors may underestimate the true variability of the coefficient estimates, potentially leading to inflated test statistics and overly narrow confidence intervals (Colin Cameron & Miller, 2015). To assess whether this concern materially affects our findings, we conduct a robust check using cluster-robust (sandwich) standard errors.

We re-estimate the OLM for each driving scenario using the cluster-robust variance estimator (CRVE). This estimator permits arbitrary within-cluster error correlation while requiring only that errors be independent across clusters. We define clusters as the nine model-language combinations (GPT-4o, DeepSeek-V3, and Llama 3.1 × English, Chinese, and French), each containing approximately 167 observations per scenario (1500 observations in total). The CRVE replaces the conventional model-based variance estimate with a sandwich-type estimator that uses the outer product of cluster-level score residuals to account for potential within-cluster dependence (Cameron et al., 2008). For each variable, we compute the SE ratio, defined as the cluster-robust standard error divided by the conventional standard error ( $SE_{Ratio} = SE_{Cluster} / SE_{Default}$ ), to quantify the degree of inflation attributable to within-cluster correlation.



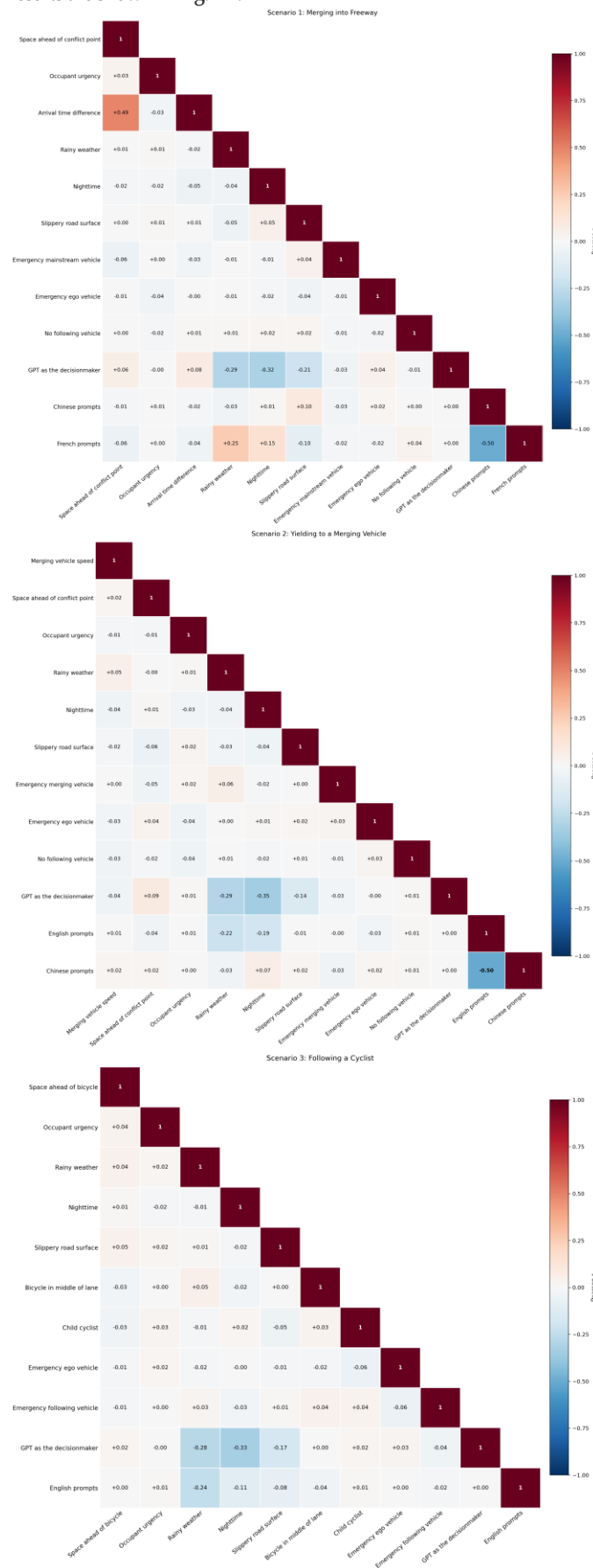
**Fig. A3.** Cluster-robust SE ratio by variable type across three driving scenarios.

**Fig. A3** presents the SE ratio for each significant variable across the three scenarios analyzed with the OLM (the pedestrian-crossing scenario is excluded due to insufficient behavioral variation; see Section 3.5). The results support the robustness of the main findings. No coefficient sign reversal is observed in any scenario. Of the 35 originally significant variables, 33 retain significance under cluster-robust inference; the two exceptions are both language indicators (English prompts in the following a cyclist scenario and Chinese prompts in the yielding to a merging vehicle scenario). The SE ratios show a clear pattern: scenario-specific contextual variables (blue in **Fig. A3**) cluster between 1.0 and 2.0, indicating minimal impact from within-group correlation, whereas model and language indicators (orange and red) exhibit higher ratios of 2.0 to 3.5. This inflation is expected, as these variables are constant within each cluster, limiting their effective between-cluster variation to only nine groups. Overall, the core variables driving our findings, occupant urgency, environmental conditions, emergency vehicle status, and spatial gap measures, remain highly significant ( $p < 0.01$ ) across all scenarios, confirming that the independence assumption does not distort the key conclusions.

### A4. Multicollinearity Diagnostics for Variables in the Ordered Logit Models

Because the OLM specifications in the main analysis include multiple variables describing spatial, temporal, and kinematic aspects of the traffic situation, it is important to verify that multicollinearity among these predictors does not compromise the stability of the coefficient estimates. To address this concern, we compute Pearson correlation matrices for all variables retained in the final OLM specification of each scenario. The

results are shown in Fig. A4.



**Fig. A4.** Pearson correlation matrices of all retained variables in the final OLM specification for the three scenarios: merging into freeway, yielding to a merging vehicle, and following a cyclist.

Across all three scenarios, the maximum absolute correlation among retained variables is 0.50 (between Chinese prompts and French prompts in the merging scenario, and between English prompts and Chinese prompts in the yielding scenario), which reflects the mechanical negative correlation between dummy-coded language indicators rather than a substantive collinearity concern. Among the contextual variables, the highest correlation is 0.49 (between space ahead of the conflict point and arrival time difference in the merging scenario); all other pairwise correlations are below 0.35. These values are well below the conventional  $|r| > 0.7$  threshold for harmful collinearity (Hair et al., 2019), confirming that multicollinearity does not pose a concern in the reported models.

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